TASK (3.1)

18/8/25

USING CLAUSES, OPERATORS AND FUNCTIONS IN QUERIES.

Aim:- To implement of DML commands using clauses, operators and functions in queries.

CLAUSES:-

→ WHERE, ORDER BY, GROUP BY, HAVING, DISTINCT

OPERATORS:-

→ Equal (=)

→ BETWEEN

→ AND

→ AND

→ OR

→ IN

```
CREATE TABLE DEPARTMENT 1(
        DEPT ID INT PRIMARY KEY,
        DEPT NAME VARCHAR(50) UNIQUE NOT NULL,
        LOCATION VARCHAR (50) NOT NULL);

CREATE TABLE STUDENT 1(
        STUDENT ID INT PRIMARY KEY,
        NAME VARCHAR (50) NOT NULL,
        AGE INT CHECK (AGE >=18),
        DEPT ID INT FOREIGN KEY REFERENCES
                        DEPARTMENT 1(DEPT ID),

    CITY VARCHAR(50) DEFAULT 'UNKNOWN';
    JOINDATE DATETIME DEFAULT GET DATE();
```

```
INSERT INTO DEPARTMENT 1 VALUES
    (1, 'CSE', 'HYDERABAD'),
    (2, 'EEE', 'MUMBAI'),
    (3, 'MECH', 'DELHI');

INSERT INTO STUDENT 1 VALUES
(101, UPPER ('rahul'), 20, 1, 'HYDERABAD');
INSERT INTO STUDENT 1 VALUES
(102, 'ANJALI', 22, 2, MUMBAI');
INSERT INTO STUDENT 1 VALUES
(103, 'KIRAN', 19, 1, 'PUNE');
INSERT INTO STUDENT 1 VALUES
(104, 'MOHITH', 23, 3, 'DELHI');
INSERT INTO STUDENT 1 VALUES
(105, 'SARA KHAN', 21, 1, HYDERABAD');
SELECT * FROM STUDENT 1;
```

S.NO	STUDENT ID	NAME	AGE	DEPT ID	CITY	JOIN DATE
1.	101	RAHUL	20	1	HYDERABAD	2025-08-26
2.	102	ANJALI	22	2	MUMBAI	2025-08-26
3.	103	KIRAN	19	1	PUNE	2025-08-26
4.	104	MOHITH	23	3	DELHI	2025-08-26
5	105	SARA KHAN	21	1	HYDERABAD	2025-08-26

```
SELECT * FROM DEPARTMENT 1;
```

S.NO	DEPT ID	DEPT NAME	LOCATION
1.	1	CSE	HYD
2.	2	EEE	MUMBAI
3.	3	MECH	DELHI

SELECT NAME , AGE
FROM STUDENT 1
WHERE AGE BETWEEN 19 AND 22;

S.NO	NAME	AGE
1	RAHUL	20
2	ANJALI	22
3	KIRAN	19
4	SARAKHAN	21

SELECT NAME, DEPT ID
FROM STUDENT 1
WHERE DEPT ID IN (1,3)
ORDER BY 1, DEPT ID DESC;

S.NO	NAME	DEPT ID
1	MOHITH	3
2	SARAKHAN	1
3	RAHUL	1
4	KIRAN	1

UPDATE STUDENT 1
SET AGE = AGE+1
WHERE DEPT ID=1 AND AGE<21;

S.NO	STUDENT ID	NAME	AGE	DEPT ID	CITY	JOIN DATE
1	101	RAHUL	21	1	HYDERABAD	2025-8-26
2	102	ANJALI	22	2	MUMBAI	2025-8-26
3	103	KIRAN	20	1	PUNE	2025-8-26
4	104	MOHITH	23	3	DELHI	2025-8-26
5	105	SARAKHAN	21	1	HYDERABAD	2025-8-26

SELECT DISTINCT CITY
FROM STUDENT 1;

S.NO	CITY
1	DELHI
2	HYDERABAD
3	MUMBAI
4	PUNE

SELECT DEPT ID, COUNT (*) AS TOTAL - STUDENTS
FROM STUDENT 1
GROUP BY DEPT ID;

S.NO	DEPT ID	TOTAL - STUDENTS
1	1	3
2	2	1
3	3	1

SELECT DEPT ID, COUNT (*) AS TOTAL - STUDENTS
FROM STUDENT 1
GROUP BY DEPT ID
HAVING COUN(*) >= 2;

S.NO	DEPT ID.	TOTAL - STUDENTS
1	1	3

VEL TECH	
EX No.	3.2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	1.5
TOTAL (20)	16.5
SIGN WITH DATE	28/8/25

RESULT:- The implementation of the Clauses, operators & functions in the queries (DDL and DML commands) all excuted sucessfully.

25/8/25 TASK (3.2) AGGREGATE FUNCTIONS

Aim:-

To ~~study &~~ implement aggregate functions (count(), sum(), Avg(), Min(), Max()) on a sample Database in my SQL.

AGGREGATE FUNCTIONS:-

They're mostly used with Grouped by to Group the rows.

→ COUNT ()
→ SUM ()
→ AVG()
→ MIN()
→ MAX ()

```
CREATE TABLE STUDENT 2 (
        ROLLNO INT PRIMARY key,
        NAME VARCHAR(50),
        AGE INT,
        DEPTID INT,
        MARKS INT);
INSERT INT STUDENT 2 VALUES
(1, "Arjun', 20, 101, 85),
(2, 'Sneha', 21, 101, 90),
(3, 'Ravi, 19, 102, 95),
(4, 'Priya', 22, 102, 95),
(5, 'kiran', 20, 101, 60),
(6, 'Anita', 23, 103, 88);
```

SELECT * FROM STUDENT 2;

	ROLL NO	NAME	AGE	DEPT ID	MARKS.
1	1	Arjun	20	101	85
2	2	Sneha	21	101	90
3	3	Ravi	19	102	70
4	4	Priya	22	102	95
5	5	kiran	20	101	60
6	6	Anita	23	103	88

SELECT DEPT ID, AVG (MARKS) AS AVG-MARKS
FROM STUDENT 2
GROUPED BY DEPT ID;

	DEPT ID	AVG_MARKS
1.	101	78
2.	102	82
3	103	88

SELECT DEPT ID, MAX (MARKS) AS TOP_MARK
FROM STUDENT 2
GROUPED BY DEPT ID;

	DEPT ID	TOP_MARK
1.	101	90
2.	102	95
3.	103	88

SELECT DEPT ID, MIN(MARKS) AS LEAST_MARK
FROM STUDENT 2
GROUP BY DEPT ID.

	DEPT ID	LAST_MARK
1.	101	60
2.	102	70
3.	103	88

```
SELECT DEPTID, COUNT(*) AS STU_COUNT
FROM STUDENT 2
GROUP BY DEPTID;
```

	DEPT ID	STU_COUNT
1.	101	3
2.	102	2
3.	103	1

VEL TECH	
EX No.	3.2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	–
TOTAL (20)	14
SIGN WITH DATE	

Result:- Implementation of all aggregate functions has been ~~Performed~~ executed Successfully. ~~on a table.~~